

Willamette Applicant Trajectory : Predicting Student College Enrollment Destination

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Abstract

This project uses three years of Willamette admissions and National Student Clearinghouse data to predict an admitted, non-enrolled applicants' enrollment destination category based on characteristics available at the time of application.

The results of most common predictors offer general insight and can be utilized by Admissions to inform enrollment strategy.



Introduction & Motivation

Problem:

A large amount of resources are poured into recruitment, through marketing and admissions outreach. Currently, recruitment and yield strategies are applied broadly, which limits the effectiveness of outreach efforts at a critical stage in the enrollment funnel. Understanding whether a student is likely to choose a private Northwest institution or a large public university would allow the team to tailor materials, financial aid messaging, and communications to what is most likely to influence a student's decision.

Questions:

- What applicant characteristics most strongly predict whether an admitted student will enroll at a private NW5 institution or a large public university?
- Are there additional meaningful enrollment destinations beyond the initial categories that arise from the data, and what applicant profiles define those groups?

Impact & stakeholders:

The beneficiaries of this project are the VP for Enrollment Management, and the broader Admissions team at Willamette University. With a well-trained model, Admissions and the Marketing team can have more efficient and effective allocation of recruitment and outreach resources.



Background & Related Work

Many institutions use enrollment-yield models to prioritize outreach effort, but these typically predict *whether* a student enrolls, not *where* they go if they don't. [1], [2] This project addresses that gap by predicting destination type, enabling more targeted, comparison-aware outreach.



Methods & Evaluation

Approach

Multinomial Logistic Regression – Serves as an interpretable baseline for the multi-class outcome that predicts an enrollment destination group.

XGBoost – Tested as a comparison model for classification performance handling NaNs internally.

Clustering – Grouping of applicant characteristics to test if cluster membership aligns with enrollment destination groups.

Validation

Models are evaluated using an 80/20 train-test split to ensure that there is adequate data to validate against.

Baseline is majority-class prediction (always guessing the most common destination).

Metrics: accuracy and macro F1 (to account for class imbalance), plus a confusion matrix for interpretability.



Results & Visualizations

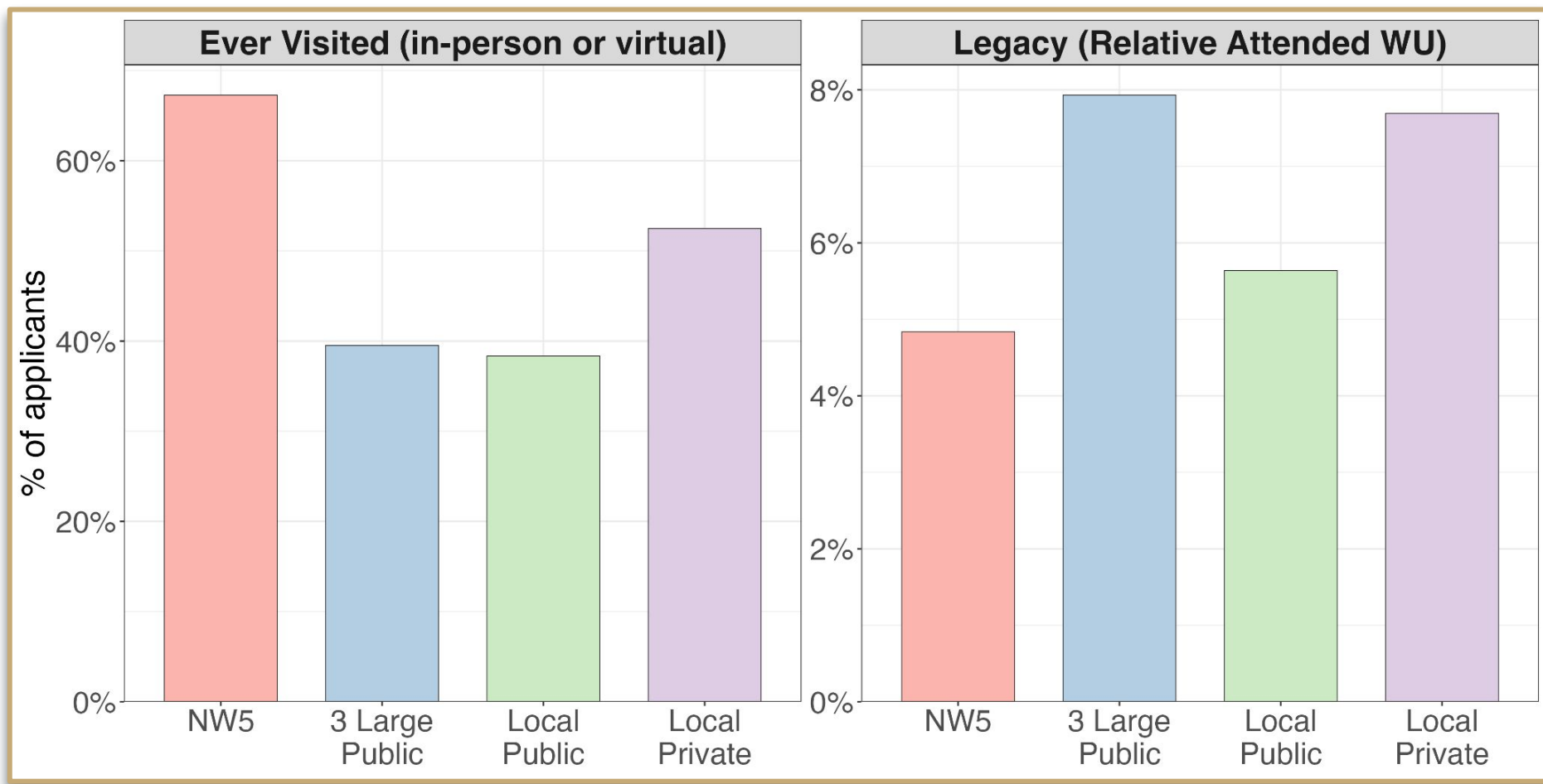


Fig 1: Engagement by Destination Group

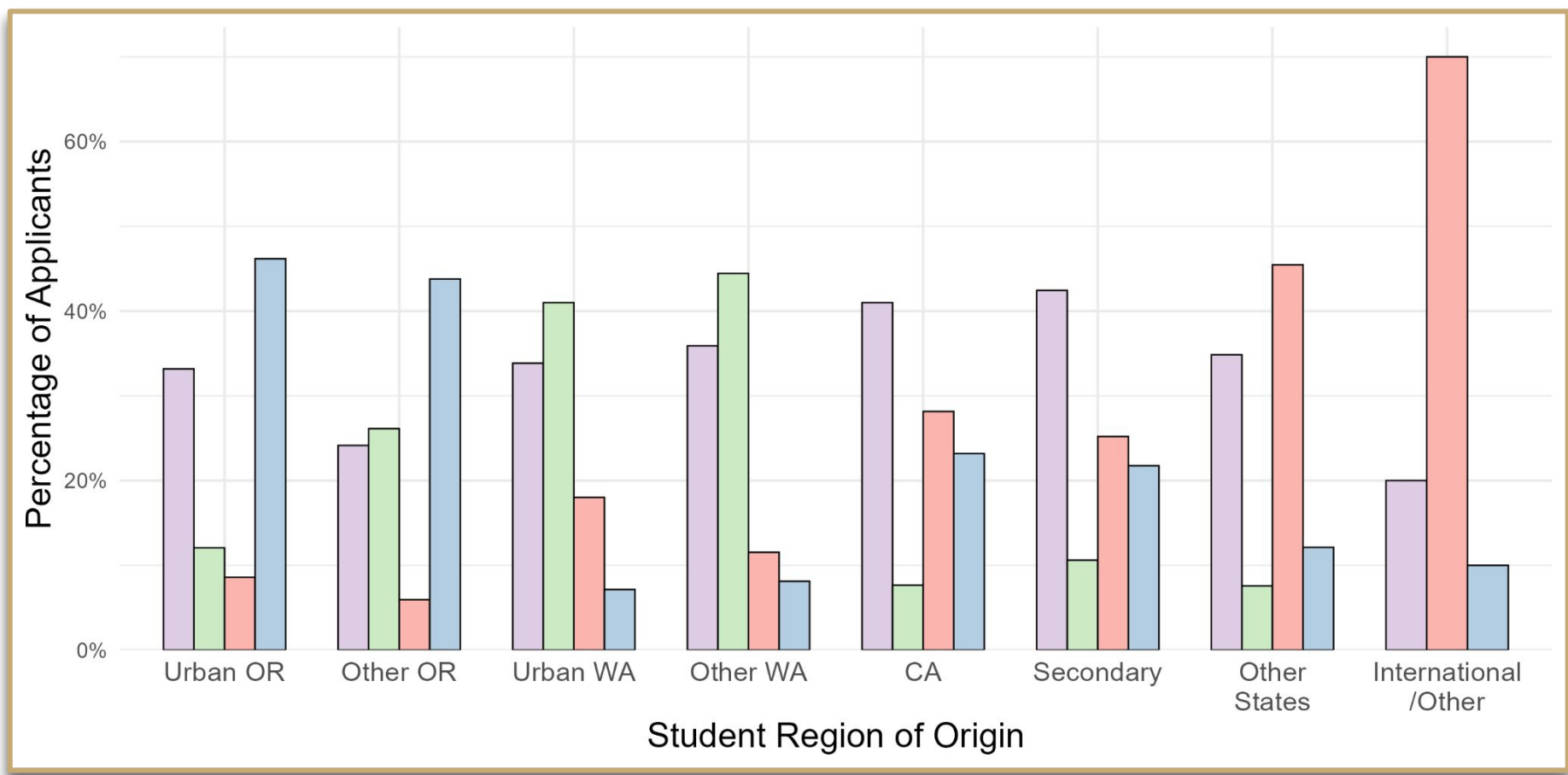


Fig 2: Destination by Student Region of Origin

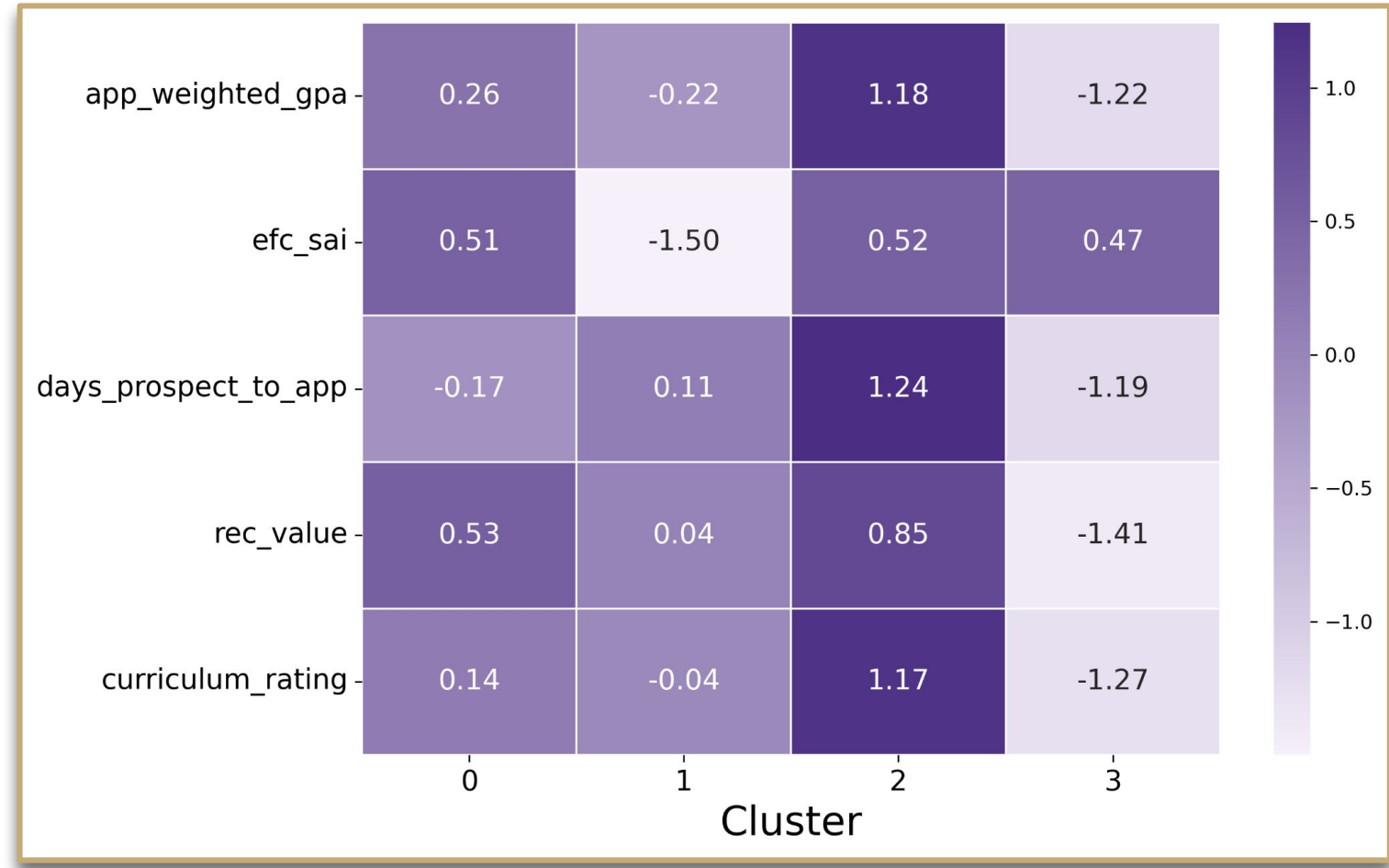


Fig 3: Cluster Profiles (Standardized)

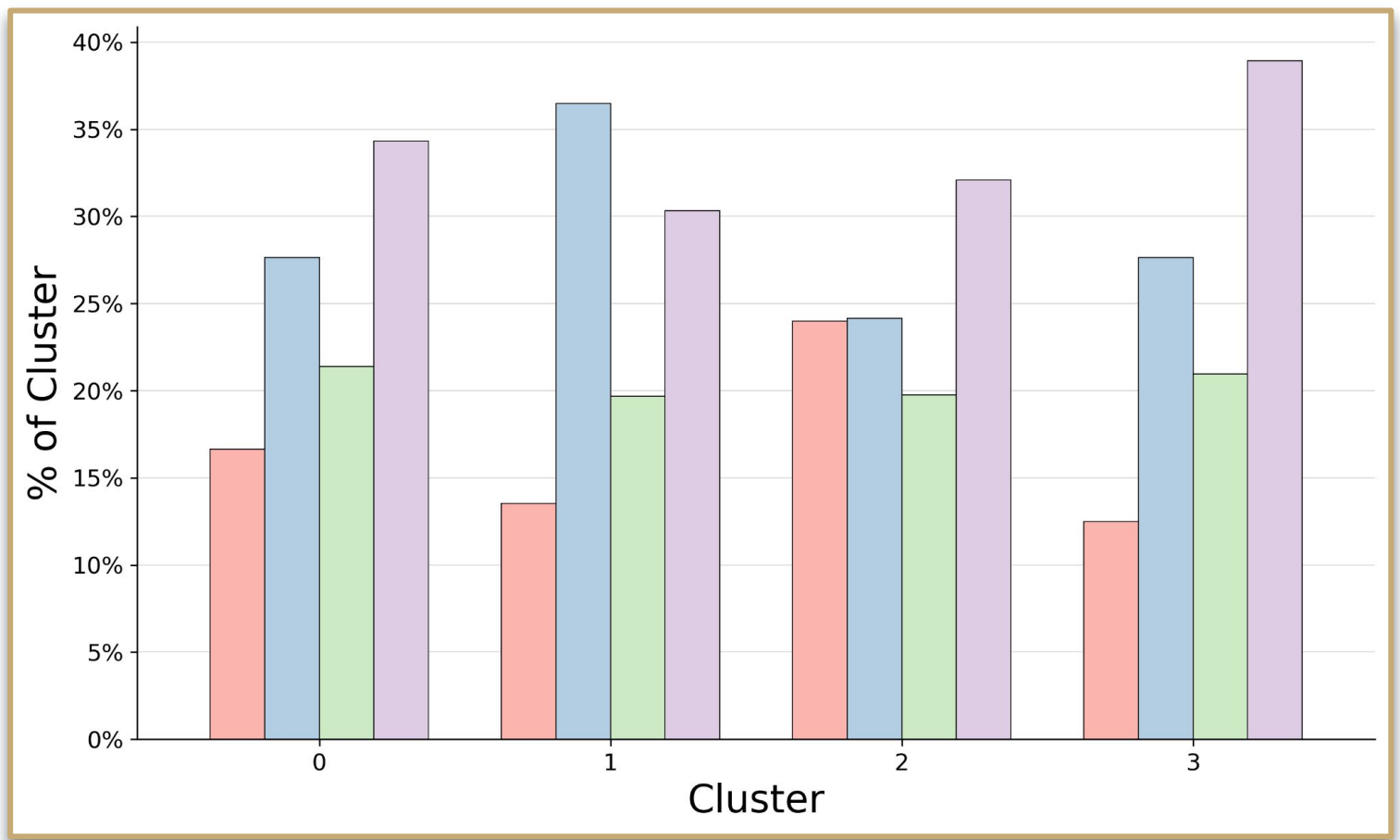


Fig 4: Destination Mix by Applicant Cluster

NW5 - 434
3 Large Public - 706
Local Public - 532
Local Private - 884

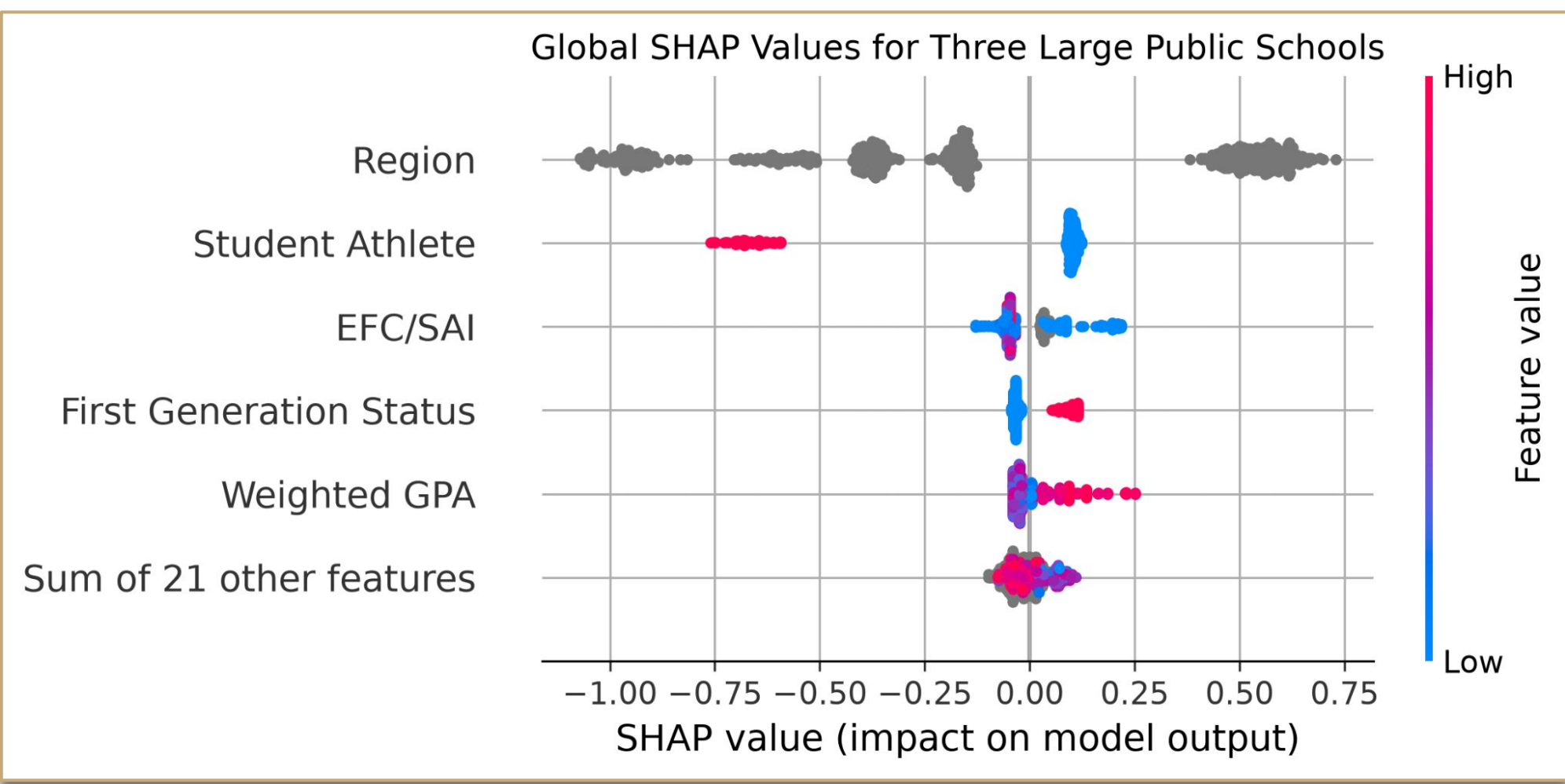
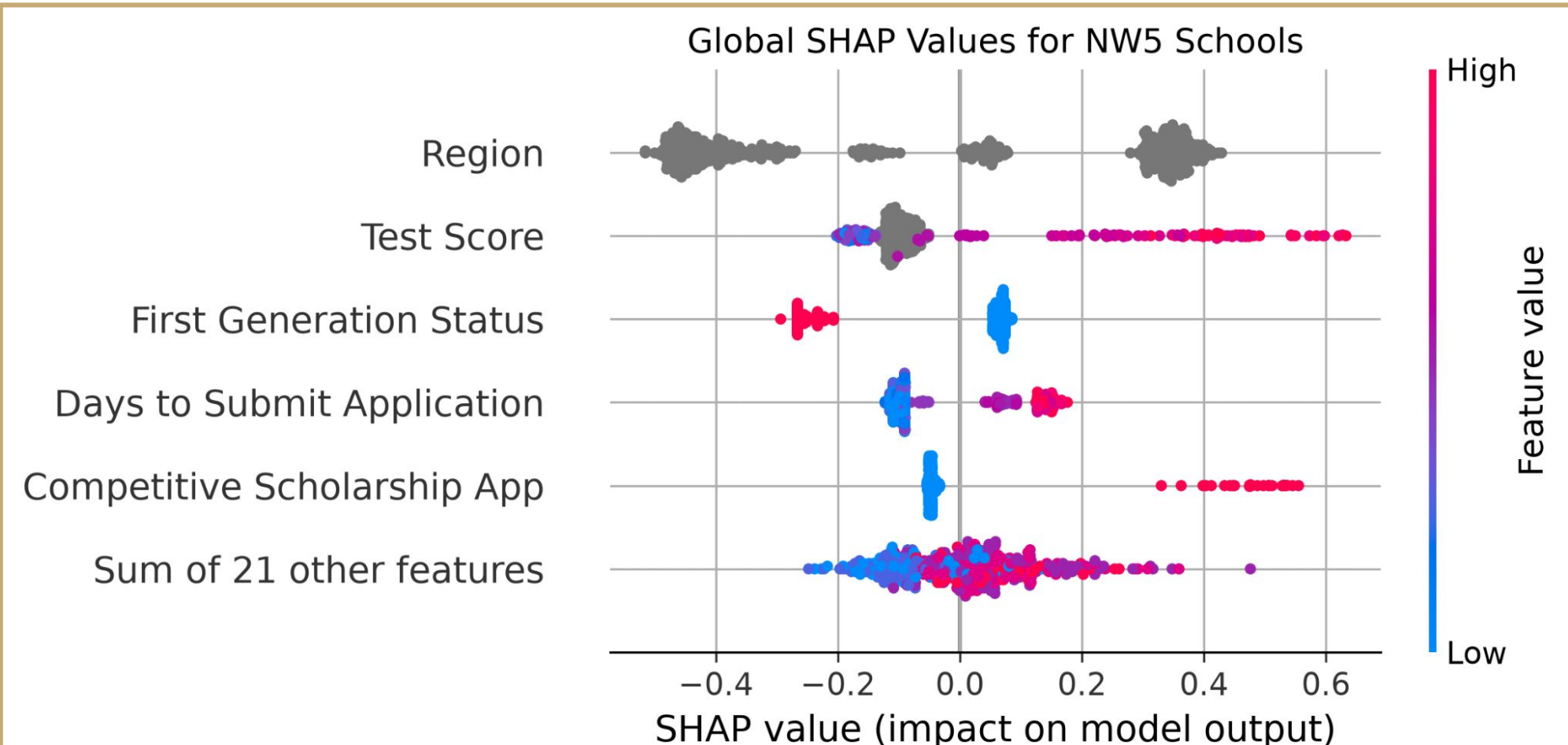


Fig 5: Feature Importance by Destination



Data, Engineering & Ethics

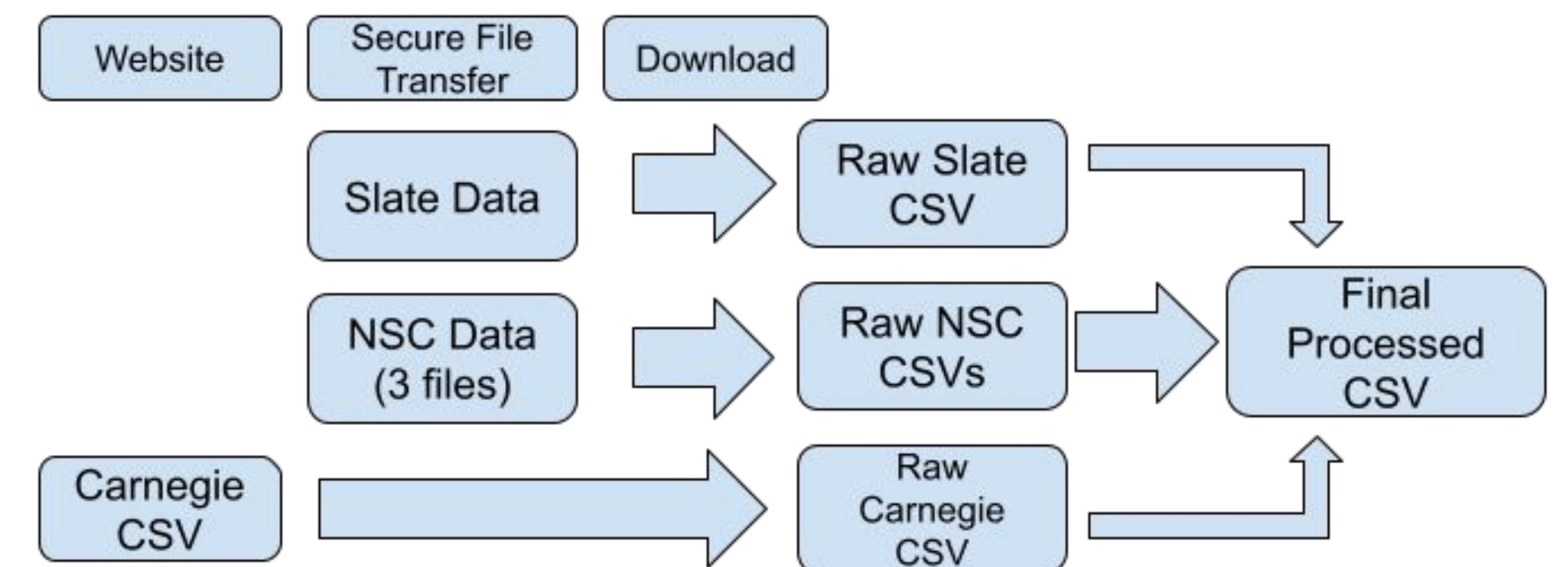
Sources & scale:

Data was drawn from Slate applicant records (9,404 rows) and three years of NSC enrollment data (2023–2025, ~6,600 rows combined), joined via applicant ID. Carnegie Classification of Institutions of Higher Education (3,928 rows) was joined via school name to create the enrollment destination target variables.

Detailed Information about the data can be found in deliverables/M2-data-summary in repo.

Pipeline: ingest → clean → transform → store (sketch below).

Ethics & responsible use: All students whose data was utilized consented to their data being used through the application process. To protect individual identities, all PII was removed from the data. To access the Slate and NSC data, team members had to sign a confidentiality agreements.



Discussion, Limitations & Conclusions

Discussion:

- Clusters represent combinations of academic strength, financial need, and application timeline behavior. However, a **Cramer's V of 0.076** indicated that cluster membership alone is not a strong standalone predictor of enrollment destination
- Region of origin** is the feature that show consistent impact on applicants' final enrollment, with students from Oregon more likely to attend a Large Public School and less likely to attend a NW5 school.
- Students are more likely to attend NW5 schools if they are not from Oregon, have higher standardized test scores, and are typically not first-generation students.
- Students who attend one of the Three Large Public Schools in Oregon tend to be from Oregon and are lower income. If an applicant is a student athlete, they are much less likely to attend a Large Public School.

Limitations & future work:

- Findings are based on three years of Willamette-specific applicant data and a comparatively small sample; results are not intended to generalize beyond this institution.
- Specific financial details were not accessed in this project due to complexity and PII concerns, which may limit the analysis of enrollment choices that are financially driven.

Conclusions:

These results offer general insight into application destination trends, but should be utilized to inform rather than replace enrollment strategy. Analysis shows that precision and recall are not high enough in the classification models to support decisions based on individual predictions alone.



Reproducibility & References

Repository: <https://github.com/Data-Science-Capstone-Project-2026/Capstone2026-Applicant-Trajectory> · Key dependencies: [Python 3.12, Packages: pandas, numpy, scikit-learn, seaborn, sklearn, kmodes, xgboost, optuna][R libraries: tidyverse, scales, tidytext, ggrridges][PostgreSQL]
References: [1] Michael Soltys, Hung Dang, Ginger Reyes, Katharine Soltys, Enrollment Predictions with Machine Learning, 2021 [2] Ahmad Slim, Don Hush, Tushar Ojha, Terry Babbitt, Predicting Student Enrollment Based on Student and College Characteristics, 2018

